

Kim Knudsen, Technical University of Denmark

Inverse Problems and Imaging

Week 2

Inverse and Ill-posed problems

Concerns the determination of a phenomenon from indirect measurements

Given forward model F and measurements y , recover the phenomena x from

$$F(x) = y.$$

Inverse and Ill-posed problems

Concerns the determination of a phenomenon from indirect measurements

Given forward model F and measurements y , recover the phenomena x from

$$F(x) = y.$$

Three questions for well-posedness according to Hadamard (1923)

- Existence
- Uniqueness
- Stability

If any fails, the problem is *ill-posed*.

Compact operators - integral operators

Many inverse problems come in the form

$$Kx = y$$

where K is a (linear) compact operator in a Hilbert space (think L^2 or $\mathbb{R}^n$).

Compact operators - integral operators

Many inverse problems come in the form

$$Kx = y$$

where K is a (linear) compact operator in a Hilbert space (think L^2 or $\mathbb{R}^n$).

Example 1: Integral operator

$$Kx(t) = \int_a^b k(t, s)x(s)ds$$

is compact in $L^2(a, b)$, e.g., if $k \in L^2((a, b)^2)$.

Example 2: A matrix $A \in \mathbb{R}^{n \times m}$ provides the compact operator

$$\mathbb{R}^m \ni x \mapsto Ax = y.$$

Model problem 0: Differentiation

Forward problem: given x , find y with $y(0) = 0$ such that $y' = x$.

Forward problem:

Model problem 0: Differentiation

Forward problem: given x , find y with $y(0) = 0$ such that $y' = x$.

Forward problem:

$$y(t) = Kx(t) = \int_0^t x(s)ds$$

Inverse problem: given a function y find the derivative $y' = x$.

Model problem 0: Differentiation

Forward problem: given x , find y with $y(0) = 0$ such that $y' = x$.

Forward problem:

$$y(t) = Kx(t) = \int_0^t x(s)ds$$

Inverse problem: given a function y find the derivative $y' = x$.

Note that

$$Kx(t) = \int_0^t x(s)ds = \int_0^1 k(t, s)x(s)ds, \quad k(s, t) = \begin{cases} 1 & s < t, \\ 0 & s > t. \end{cases}$$

Model problem 1: Abel integral

From Computerized Tomography in radially symmetric case:
Abel integral operator

$$Kx(t) = \int_0^1 k(t,s)x(s)ds, \quad t > 0, \quad k(t,s) = \begin{cases} \frac{1}{\sqrt{(t-s)}}, & s < t \\ 0, & s > t. \end{cases}$$

Inverse problem: from $y = Kx$, find x .

Model problem 2: Backwards Heat Equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \quad 0 < x < \pi, \quad t > 0$$

Boundary conditions: $u(0, t) = u(\pi, t) = 0$

Initial condition: $u(x, 0) = f(x)$.

Model problem 2: Backwards Heat Equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \quad 0 < x < \pi, \quad t > 0$$

Boundary conditions: $u(0, t) = u(\pi, t) = 0$

Initial condition: $u(x, 0) = f(x)$.

Forward problem: from f to terminal temperature $g = u(\cdot, T)$

$$g(x) = Kf(x) = \int_0^\pi k(x, y)f(y) dy, \quad k(x, y) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nx) \sin(ny).$$

Inverse problem: Go back in time from g to f .

Model problem 2: Backwards Heat Equation

With x the initial heat distribution, y the terminal, s, t space variables,

$$y(t) = Kx(t) = \int_0^\pi k(t, s)x(s) \, ds$$

with

$$k(t, s) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nt) \sin(ns).$$

Inverse problem: Go back in time from y to x .

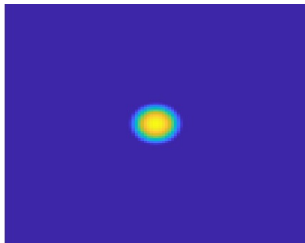
Model problem 3: Image deblurring

Deblurring of images due to lens out of focus, imperfections...

 f  $g = Kf$

$$Kf(y) = \int \phi(x - y)f(y)dy$$

Convolution $Kf(x) = \int \phi(x - y)f(y)dy$



Discretization of integral equation

Operator equation vs matrix equation

$$Kx(t) = \int_a^b k(t, s)x(s)ds = y(t) \quad \rightarrow \quad Ax = y$$

Quadrature + collocation (Nyström):

$$\int_a^b k(t, s)x(s)ds \approx \sum_{j=1}^J w_j k(t, s_j)x(s_j),$$
$$\sum_{j=1}^m w_j k(t_i, s_j)x(s_j) = y(t_i), \quad i = 1, 2, \dots, n.$$

That is $Ax = y$ with

$A_{ij} = w_j k(t_i, s_j)$, $y_i = y(t_i)$, x_j : approximation of samples $x(s_j)$.

Discretization of integral equation

$$Kx(t) = \int_a^b k(t, s)x(s)ds = y(t) \quad \rightarrow \quad Ax = b$$

Expansion method (Galerkin):

Consider for two basis (for finite dimensional subspaces of $L^2(a, b)$) $\{\phi_j\}$ and $\{\psi_i\}$ and approximate

$$x \in \text{span}(\phi_1, \phi_2, \dots, \phi_m), \quad y \in \text{span}(\psi_1, \psi_2, \dots, \psi_n).$$

Requiring the *residual* $Kx - y$ to be orthogonal to $\text{span}(\psi_1, \psi_2, \dots, \psi_n)$

$$A_{ij} = \int \int \psi_i(t)k(t, s)\phi_j(s) dsdt, \quad y_i = \int \psi_i(t)y(t) ds,$$

x_j : expansion coefficients of x .

Role of exact solutions

Pairs of $x(s)$, $y(t) = Kx(t)$ allow us to

- 1 Verify code for forward solution K
- 2 Compare computed solution to inverse problem to analytic solution
- 1 Differentiation
- 2 Abel (see exercise)
- 3 Heat (basis functions)
- 4 Blurring??